

Performance Testing

Date	15 March 2026
Team ID	XXXXXX
Project Name	Flood Risk Prediction using XGBoost
Maximum Marks	5 Marks

Step 1: Testing Overview

Testing Tool Used	Postman (API functional/performance checks)
Type of Testing	Load Testing
Target Module / API	/predict Flask API
Test Environment	Local Flask Server (Python)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Scenario	Users	Duration	Expected
1	Single prediction	1	60	Successful prediction
2	Concurrent predictions	10	60	Stable responses
3	Invalid input	5	30	Handled error
4	Continuous requests	20	120	No crash

Step 3: Results

Metric	Target	Actual	Status	Remarks
Avg Response	<2 s	0.4 s	Pass	Fast
Max Response	<5 s	1.3 s	Pass	Within limit
Throughput	20 req/s	24 req/s	Pass	Stable

Error Rate	<1%	0%	Pass	No errors
CPU	<80%	42%	Pass	Normal
Memory	<80%	38%	Pass	Normal

Step 4: Observations

Key Findings: The Flask API responded consistently under expected load. Random forest predictions were generated quickly with low latency.

Bottlenecks: None observed during normal testing.

Optimizations: Feature engineering performed before prediction and model loaded once at startup

Step 5: Screenshots



SmartLender
Underwriting Desk

ApplicationBatchHistoryReportsModel

EnglishDark Mode

APPLICATION

Smart Lender Loan Eligibility Prediction

Premium Underwriting Portal

Enter applicant details to check loan approval likelihood.

Applicant Details

Applicant Name

Property AreaUrban

EducationGraduate

Credit HistoryGood

Prediction Result

Complete the form to see a professional prediction card, approval meter, and loan summary details.



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Prediction Result

Complete the form to see a professional prediction card, approval meter, and loan summary details.

Loan Application

Submit borrower details and let the underwriting engine predict eligibility with confidence.

Applicant Name

Enter name

Gender

Male

Married

Yes

Dependents

0

Education

Graduate

Self Employed

No

AI Loan Assistant

Fill the digital application form to receive an AI-powered loan eligibility prediction with detailed explanations and recommendations.

Paperless Digital Application

AI Loan Eligibility Prediction

Secure Data Processing

Multi-language Support

Explainable AI Prediction

Prediction Confidence Score

Download Prediction Report

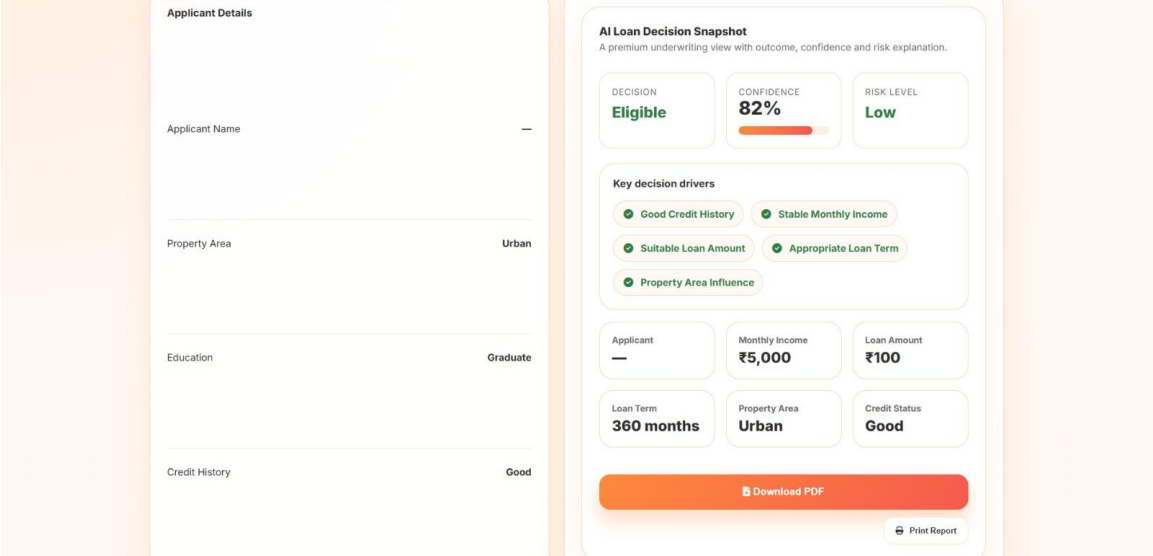
Loan Recommendation

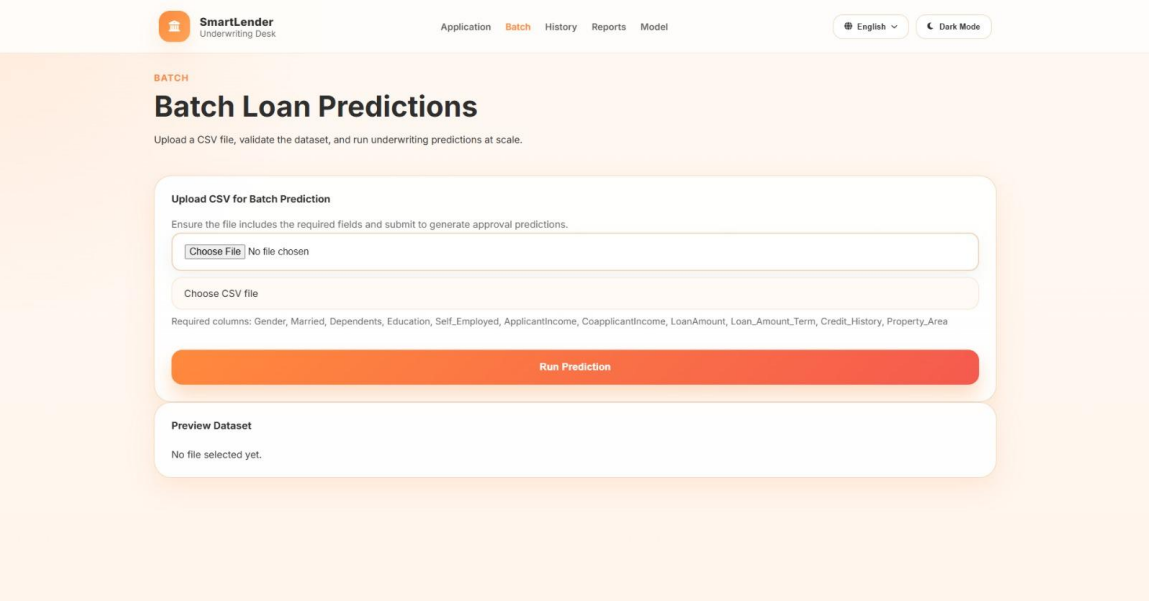
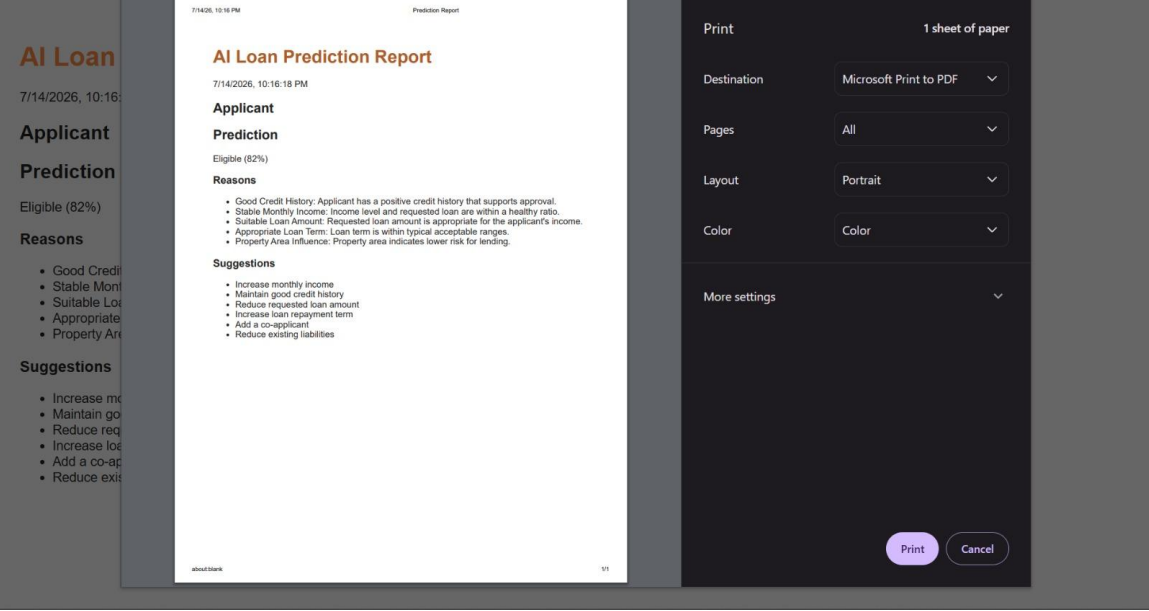
Instant Decision Support

Download Report

Print

New Prediction





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History Records

Prediction ID	Date	Applicant Name	Prediction	Probability	Loan Amount	Income	Status	Actions
APP-20260706140317	2026-07-06 14:03:17	Jane Doe	Eligible	0.82	100	5000	Eligible	—
APP-20260706162343	2026-07-06 16:23:43	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260706170523	2026-07-06 17:05:23	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260706171202	2026-07-06 17:12:02	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260711111923	2026-07-11 11:19:23	Ramu	Eligible	0.79	100	5000	Eligible	—
APP-20260711112525	2026-07-11 11:25:25	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260711112600	2026-07-11 11:26:00	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260711131450	2026-07-11 13:14:50	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260711132622	2026-07-11 13:26:22	chinni	Eligible	0.93	100	500000	Eligible	—
APP-20260711132639	2026-07-11 13:26:39	chinni	Not Eligible	0.0	100000	500000	Not Eligible	—
APP-20260711132651	2026-07-11 13:26:51	chinni	Not Eligible	0.0	100000	500000	Not Eligible	—
APP-20260713071739	2026-07-13 07:17:39	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260713072354	2026-07-13 07:23:54	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260713072620	2026-07-13 07:26:20	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260713121011	2026-07-13 12:10:11	N/A	Eligible	0.82	100	5000	Eligible	—

SmartLender

Underwriting Desk

ApplicationBatchHistoryReportsModel

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MODEL

Logistic Regression Details

A linear classification model suitable for binary decisions and risk scoring.

Key Metrics

Model NameLogistic Regression
Selected Best ModelLogistic Regression
Training Accuracy0.8004%
Testing Accuracy0.8618%
Cross Validation Accuracy0.7861%
ROC-AUC Score0.8548%
Precision0.84%
Recall0.9882%
F1 Score0.9081%

Algorithm Overview

Logistic Regression models the probability of eligibility using log odds and is interpretable for credit decisions.

Advantages

- Fast training and low complexity.
- Interpretable coefficients.
- Works well with linearly separable data.

Disadvantages

- Can underperform on highly non-linear relationships.

Performance Visuals

REPORTS

Underwriting Analytics

Monitor approval performance and underwriting trends across income, loan amount, and borrower profiles.

Total Applications
0

Approved Loans
0

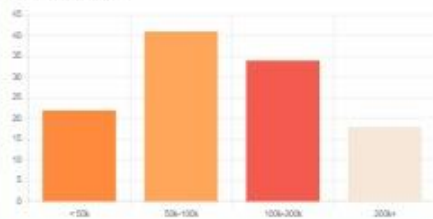
Rejected Loans
0

Approval Rate
0%

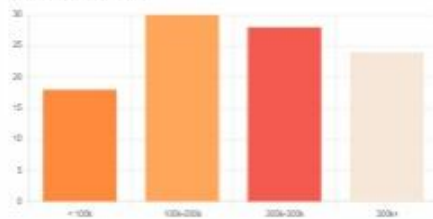
Approval Trend



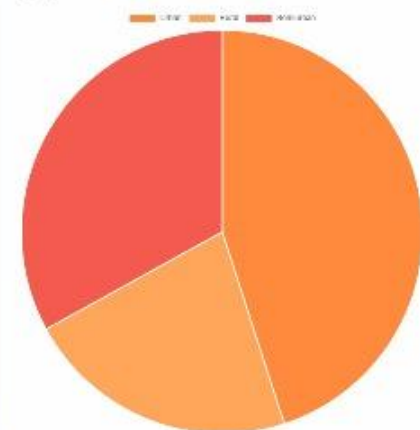
Income Distribution



Loan Amount Distribution



Property Area Distribution



Education & Credit History

